

Applications to the system of linear equations

Example 1

Ali and Sara are shopping for chocolate bars.

- Ali said to sara, “If I add half my amount to yours, it will be enough to buy two chocolate bars.”
- Sara asks, “If I add half my amount to yours, how many can we buy?” Ali replies, “One chocolate bar.”

How much money did Ali have?

Solution:

Let

a = Amount of Ali he has

s = Amount of Sara she has

c = Cost of chocolate

Then the system of equations can be written as

$$\begin{cases} \frac{1}{2}a + s = 2c & \text{--- (1)} \\ a + \frac{1}{2}s = c & \text{--- (2)} \end{cases}$$

Multiplying equation (1) by “2” both sides and similarly Multiplying equation (2) by “2” both sides, we get

$$\begin{cases} a + 2s = 4c & \text{--- (3)} \\ 2a + s = 2c & \text{--- (4)} \end{cases}$$

The augmented matrix of the above system of equations is

$$\begin{bmatrix} 1 & 2 & 4c \\ 2 & 1 & 2c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4c \\ 0 & -3 & -6c \end{bmatrix} \quad \sim \sim \sim \sim \sim \sim \sim \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 4c \\ 0 & 1 & 2c \end{bmatrix} \sim -\frac{1}{3}R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2c \end{bmatrix} \sim R_1 - 2R_2$$

Hence, the solution of the given system is $(a, s) = (0, 2c)$.

It means **Ali has no amount**.

Example 2

Three **Alto**, two **Suzuki**, and four **City** can be rented for \$ **106** per day. At the same rates two Alto, four Suzuki, and three City cost \$ **107** per day, whereas four Alto, three Suzuki, and two City cost \$ **102** per day. Find the rental rates for all three kinds of cars?

Solution:

Let

a = Rental rate of Alto Car per day

s = Rental rate of Suzuki Car per day

c = Rental rate of City Car per day

Then, we can form the system of equations as

$$3a + 2s + 4c = 106$$

$$2a + 4s + 3c = 107$$

$$4a + 3s + 2c = 102$$

The augmented matrix of the above system is

$$\begin{bmatrix} 3 & 2 & 4 & : & 106 \\ 2 & 4 & 3 & : & 107 \\ 4 & 3 & 2 & : & 102 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 1 & : & -1 \\ 2 & 4 & 3 & : & 107 \\ 4 & 3 & 2 & : & 102 \end{bmatrix} \sim R_1 - R_2$$

Example 3

A restaurant owner plans to use x tables seating 4 customers, y tables seating 6 customers and z tables seating 8 customers, for a total 20 tables. When fully occupied, the tables seat 108 customers. If only half of the x tables, half of the y tables and one-fourth of the z tables are used, each fully occupied, then 46 customers will be seated. Find x , y , and z .

Solution:

Let us consider

- x denotes the number of tables seating 4 customers each
- y denotes the number of tables seating 6 customers each
- z denotes the number of tables seating 8 customers each

From the given information, we can write the system of equations as:

The equation for total number of tables is

$$x + y + z = 20$$

The equation for total number of customers is

$$4x + 6y + 8z = 108$$

The equation for half of the x tables, half of the y tables and one-fourth of the z tables are being used is

$$4\left(\frac{x}{2}\right) + 6\left(\frac{y}{2}\right) + 8\left(\frac{z}{4}\right) = 46$$

Simplifying the system, we have

$$\begin{cases} x + y + z = 20 \\ 2x + 3y + 4z = 54 \\ 2x + 3y + 2z = 46 \end{cases}$$

•
•
•

By solving the system, we get: $x = 10$, $y = 6$ and $z = 4$.

Hence

$$(x, y, z) = (10, 6, 4)$$

Example 4

Suppose I have **32** bills in my wallet, in the denominations (values) of US \$ **1**, \$ **5** and \$ **10**, worth \$ **100** in total. How many do I have of each denomination?

Solution:

Let

- x = No. of \$ **1** bills,
- y = No. of \$ **5** bills,
- z = No. of \$ **10** bills.

Then the system of equations will become

$$\begin{cases} x + y + z = 32 \longrightarrow (1) \\ x + 5y + 10z = 100 \longrightarrow (2) \end{cases}$$

The augmented matrix of the above given system is

$$\begin{aligned} & \begin{bmatrix} 1 & 1 & 1 & : & 32 \\ 1 & 5 & 10 & : & 100 \end{bmatrix} \\ & \begin{bmatrix} 1 & 1 & 1 & : & 32 \\ 0 & 4 & 9 & : & 68 \end{bmatrix} \quad \sim \sim \sim \sim \sim \sim \sim \quad R_2 - R_1 \\ & \begin{bmatrix} 1 & 1 & 1 & : & 32 \\ 0 & 1 & \frac{9}{4} & : & 17 \end{bmatrix} \quad \sim \sim \sim \sim \sim \sim \sim \quad \frac{1}{4}R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{5}{4} & 15 \\ 0 & 1 & \frac{9}{4} & 17 \end{array} \right] \sim \sim \sim \sim \sim \sim \sim \sim R_1 - R_2$$

Here x, y, z should be positive integers. Now, by using **back substitution**, we get

$$\begin{cases} 0x + 1y + 94z = 17 & - & - & - & - & (3) \\ 1x + 0y - 54z = 15 & - & - & - & - & (4) \end{cases}$$

Since the given system is **under-determined system**, therefore, let

$$z = t - - - - - (5)$$

Then equation (3) will give the value of y is

$$y = 17 - 94t - - - - - (6)$$

By putting values of equation (5) and (6) in equation (4), we get the value of x is

$$x = 15 + 54t - - - - - (6)$$

Since the variables x, y and z represents the denominations of the bills, that cannot be negative. Therefore, it must be positive. This implies that the value of t would be

$$t = 4, 8, 12, 16 \dots \dots \dots$$

For the given situation, put $t = 4$, in equation (5), (6) and (7) to get the values of x, y and z . That is,

$$z = t = 4$$

$$y = 17 - 94(4) = 8$$

$$x = 15 + 54(4) = 20$$

So, I have

- **20 bills** of amount \$ **1**
- **8 bills** of amount \$ **5**
- **4 bills** of amount \$ **10**.

Example 5

Ali is getting some flowers for his office. Being of a precise analytical mind, he plans to spend exactly \$ **24** on a bunch of exactly two dozen flowers. At the flower market they have **lilies** (\$ **3** each), **roses** (\$ **2** each), and **daisies** (\$ **0.50** each). Ali loves lilies, what is he to do?

Solution:

Let

- x = No. of **Lilies** Ali buys,
- y = No. of **Roses** Ali buys,
- z = No. of **Daisies** Ali buys.

Then the system of equations will become

$$\begin{cases} 3x + 2y + 0.5z = 24 & \text{--- (1)} \\ x + y + z = 24 & \text{--- (2)} \end{cases}$$

We can simplify equation (1) by multiplying “2” both sides, and the system becomes

$$\begin{cases} 6x + 4y + z = 48 & \text{--- (3)} \\ x + y + z = 24 & \text{--- (4)} \end{cases}$$

The augmented matrix of the above given system is

$$\left[\begin{array}{ccc|c} 6 & 4 & 1 & 48 \\ 1 & 1 & 1 & 24 \end{array} \right]$$

•
•
•

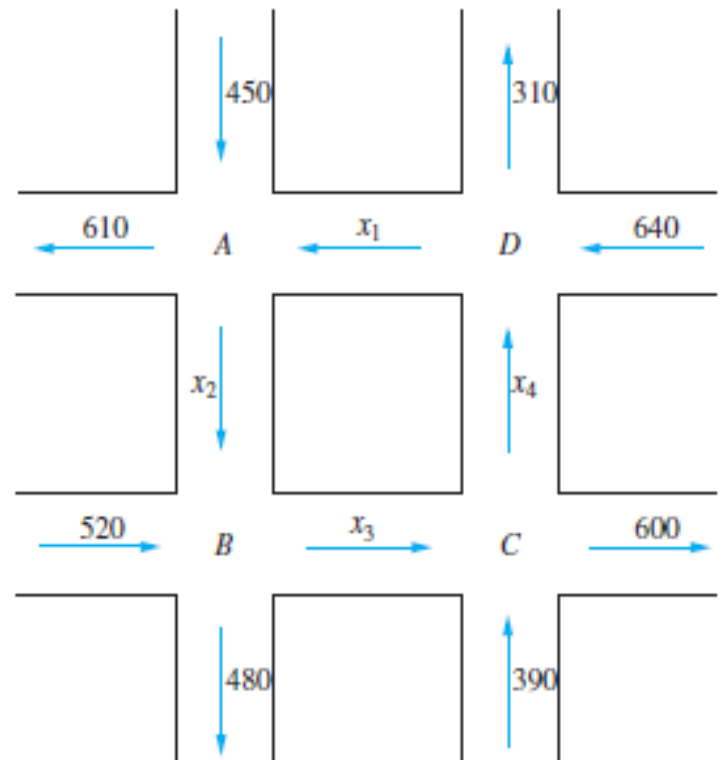
Solve it by yourself

Answer:

$$(x, y, z) = \left(\frac{3}{2}t - 24, \quad 48 - \frac{5}{2}t, \quad t \right)$$

Example 6

In the downtown section of a certain city, two sets of one-way streets intersect as shown in Figure below. The average hourly volume of traffic entering and leaving this section during rush hour is given in the diagram. At each intersection, the number of automobiles entering must be the same as the number leaving.



- (a) Determine the amount of traffic between each of the four intersections.
- (b) The amount of traffic between intersections C and D averages 200 automobiles. Then find the amount of traffic for other intersections.

Solution:

From the figure, we have four (4) intersections *A*, *B*, *C* and *D*.

Intersection	Automobiles Entering at intersection	Automobiles Leaving at intersection
<i>A</i>	$x_1 + 450$	$x_2 + 610$
<i>B</i>	$x_2 + 520$	$x_3 + 480$
<i>C</i>	$x_3 + 390$	$x_4 + 600$
<i>D</i>	$x_4 + 640$	$x_1 + 310$

Since given that, at each intersection, the number of automobiles **entering must be the same as the number leaving**. Therefore

Intersection A: $x_1 + 450 = x_2 + 610$

Intersection B: $x_2 + 520 = x_3 + 480$

Intersection C: $x_3 + 390 = x_4 + 600$

Intersection D: $x_4 + 640 = x_1 + 310$

Now, rearranging the equations to convert it into standard form, we have

Intersection A: $x_1 - x_2 = 610 - 450 \Rightarrow x_1 - x_2 = 160$

Intersection B: $x_2 - x_3 = 480 - 520 \Rightarrow x_2 - x_3 = -40$

Intersection C: $x_3 - x_4 = 600 - 390 \Rightarrow x_3 - x_4 = 210$

Intersection D: $x_4 - x_1 = 310 - 640 \Rightarrow x_4 - x_1 = -330$

Multiplying the equation of Intersection D by (-1) , we get $x_1 - x_4 = 330$

By simplifying, we get the system of equations given as

Intersection A: $x_1 - x_2 = 160 \text{ --- (1)}$

Intersection B: $x_2 - x_3 = -40 \text{ --- (2)}$

Intersection C: $x_3 - x_4 = 210 \text{ --- (3)}$

Intersection D: $x_1 - x_4 = 330 \text{ --- (4)}$

The augmented matrix of the above system of equations is given as

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 160 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 1 & 0 & 0 & -1 & 330 \end{array} \right]$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 : 160 \\ 0 & 1 & -1 & 0 : -40 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 1 & 0 & -1 : 170 \end{bmatrix} \xrightarrow{\text{-----}} R_4 \Rightarrow R_4 - R_1$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 : 160 \\ 0 & 1 & -1 & 0 : -40 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 0 & 1 & -1 : 210 \end{bmatrix} \xrightarrow{\text{-----}} R_4 \Rightarrow R_4 - R_2$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 : 160 \\ 0 & 1 & -1 & 0 : -40 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 0 & 0 & 0 : 0 \end{bmatrix} \xrightarrow{\text{-----}} R_4 \Rightarrow R_4 - R_3$$

(Row Echelon Form)

Now,

$$\begin{bmatrix} 1 & 0 & -1 & 0 : 120 \\ 0 & 1 & -1 & 0 : -40 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 0 & 0 & 0 : 0 \end{bmatrix} \xrightarrow{\text{-----}} R_1 \Rightarrow R_1 + R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 : 330 \\ 0 & 1 & -1 & 0 : -40 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 0 & 0 & 0 : 0 \end{bmatrix} \xrightarrow{\text{-----}} R_1 \Rightarrow R_1 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 : 330 \\ 0 & 1 & 0 & -1 : 170 \\ 0 & 0 & 1 & -1 : 210 \\ 0 & 0 & 0 & 0 : 0 \end{bmatrix} \xrightarrow{\text{-----}} R_2 \Rightarrow R_2 + R_3$$

(Reduced Row Echelon Form)

Since, last row of the matrix is a complete zero row. That is

$$0x + 0y + 0z = 0 \Rightarrow 0 = 0$$

which means that one variable will be treated as free variable. Therefore, let

$$x_4 = t \text{ (Free Variable)}$$

Then, by simplifying, we get the solution is

$$x_4 = t$$

$$x_3 = t + 210$$

$$x_2 = t + 170$$

$$x_1 = t + 330$$

Hence,

$$(x_1, x_2, x_3, x_4) = (t + 330, t + 170, t + 210, t)$$

(b) Given that, the amount of traffic between intersections *C* and *D* averages 200 automobiles. That is,

$$x_4 = t = 200$$

Hence, the amount of traffic for other intersections is

$$x_3 = t + 210 = 200 + 210 = 410$$

$$x_2 = t + 170 = 200 + 170 = 370$$

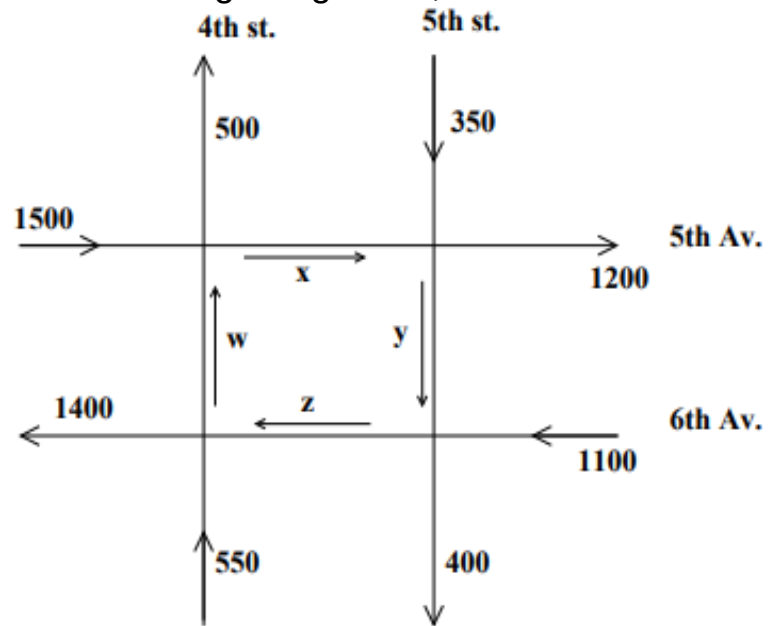
$$x_1 = t + 330 = 200 + 330 = 530$$

Hence,

$$(x_1, x_2, x_3, x_4) = (530, 370, 410, 200)$$

Question 6

The accompanying figure shows the flow of downtown traffic in a certain city during rush hours on a typical weekday. The arrows indicate the direction of traffic flow on each one-way road, and the average number of vehicles entering and leaving each intersection per hour appears beside each road. Fifth and Sixth Avenues can handle up to **2000** vehicles per hour without causing congestion, whereas the maximum capacity of each of the two streets is **1000** vehicles per hour. The flow of traffic is controlled by traffic lights installed at each of the four intersections.



(a) Set up the system of equations that would model this problem.

(b) Solve the system of equations and write the answer in parametric form. Place restrictions on the parameter.

(c) Find two possible flow patterns that would ensure that there is no traffic congestion.

The system of equations was generated by the following thought: what goes into an intersection must come out of an intersection.

Definition of variables:

x = average number of cars on 5th Av.

y = average number of cars on 5th St.

z = average number of cars on 6th Av.

w = average number of cars on 4th St.

System of equations:

$$x - w = 1000$$

$$x - y = 850$$

$$y - z = -700$$

$$z - w = 850$$

Initial matrix

final matrix

Parametric Solution:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 1000 \\ 1 & -1 & 0 & 0 & 850 \\ 0 & 1 & -1 & 0 & -700 \\ 0 & 0 & 1 & -1 & 850 \end{array} \right] \xrightarrow{\text{rref}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 1000 \\ 0 & 1 & 0 & -1 & 150 \\ 0 & 0 & 1 & -1 & 850 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x = 1000 + w$$

$$y = 150 + w$$

$$z = 850 + w$$

$$w = \text{any number}$$

Since this is a word problem, we can place restrictions on the parameter. We know that all of the variables must be greater than or equal to zero.

$$x \geq 0$$

$$y \geq 0$$

$$z \geq 0$$

$$w \geq 0$$

$$100 + w \geq 0$$

$$150 + w \geq 0$$

$$850 + w \geq 0$$

$$w \geq -1000$$

$$w \geq -150$$

$$w \geq -850$$

In addition we know that x and z can not be any larger than 2000 and y and w can not be any larger than 1000.

$$x \leq 2000$$

$$y \leq 1000$$

$$z \leq 2000$$

$$w \leq 1000$$

$$100 + w \leq 2000$$

$$150 + w \leq 1000$$

$$850 + w \leq 2000$$

$$w \leq 1000$$

$$w \leq 850$$

$$w \leq 1150$$

When all of these restrictions are considered, we find that in order to have a valid solution, we need the restriction on the parameter to be $\mathbf{0 \leq w \leq 850}$.

Practice Questions for Students:

Question 1: Students are buying books for the new semester. Asma buys the linear algebra book and the differential equation book for \$ **178**. Aiman, who is buying books for herself and her friend, spends \$ **319** on two linear algebra books, one differential equation book, and one educational psychology book. Sara buys the educational psychology book and the differential equation book for \$ **147** in total. How much does each book cost?

Question 2: A soap manufacturer wants to spend **60 Lac** rupees on radio, magazine, and TV advertising. If he spends as much on TV advertisement as on magazines and radio together, and the amount spend on magazines and TV combined equals 5 times that spent on radio, what is the amount to be spent on each type of advertising?

Question 3: **Three** merchants find a purse lying in the road. One merchant says “If I keep the purse, I shall have twice as much money as the two of you together”. “Give me the purse and I shall have three times as much as the two of you together”, said the second merchant. The third merchant said, “I shall be much better off than either of you if I keep the purse, I shall have five times as much as the two of you together.” If there are **60** coins (of equal value) in the purse, how much money does each merchant have?

Question 4: The new animated feature film is now playing **3** times a day in DHA Cinema. One day, there were **20** adults, **43** children and **10** senior citizens and theater made \$ **614**. At the next showing there were **24** adults, **59** children and **20** senior citizens with the theater making it \$ **852** in ticket sale. At the last showing theater made \$ **405** with **13** adults, **30** children and **5** senior citizens. How much are the tickets price at the movie theater?

Answer: Tickets prices of adults, children, senior citizen **10,8,7** dollars respectively. $(x, y, z) = (10, 8, 7)$

Question 5: A duck cost four coin, a hen cost **3** coin, and a sparrow cost one coin. Ali buys **50** birds for **100** coins. How many birds of each kind can he buy? Make a system of linear equations and find their solution.